The Dark Mirror:
Dual-Use Risks of Number-Theoretic Engineering from the Cathedral Project

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Abstract

The companion papers *What Could Be Built* and *The Prime Number Grid* propose engineering applications of the mathematical structures discovered during the Cathedral project. Responsible disclosure requires examining the same technologies from an adversarial perspective. This paper identifies five dual-use risks: (1) precision harmonic injection attacks on power grids, (2) resonance-targeted industrial sabotage via arithmetic vibration analysis, (3) adversarial use of verified instability certificates for cascade failure engineering, (4) broadband stealth and jamming from prime-spaced antenna arrays, and (5) covert RF surveillance via arithmetic energy harvesting. For each risk, we assess the severity, the novelty (whether the attack vector is genuinely new), the current state of defenses, and recommended mitigations. We conclude that the mathematical structures in the Cathedral do not create fundamentally new attack capabilities, but could make existing attack vectors more *precise* and harder to detect. We recommend specific defensive measures and argue that publishing this analysis openly is the responsible path: attackers already have access to these mathematical tools; defenders need to understand them too. This paper exists because the authors believe that *thinking about misuse before deployment* is a moral obligation of anyone who builds powerful tools.

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1 Why This Paper Exists

Every powerful tool has a dual use. The same chemistry that produces fertilizer produces explosives. The same physics that enables nuclear medicine enables nuclear weapons. The same number theory that could improve power grid monitoring could, in principle, be turned against the grid.

The companion papers propose six positive energy applications. This paper examines what happens when each is reflected through an adversarial lens. We do not provide implementation details for any attack—we describe the *category* of risk and the *structural* vulnerability, not the recipe.

1.1 Responsible Disclosure Rationale

We publish this analysis openly for three reasons:

1. **Attackers already know the math.** The Möbius function, Fourier analysis, spectral theory, and variational methods are textbook material. Anyone with a graduate education in mathematics or engineering already has access to these tools. Suppressing this paper would not deny attackers any capability.

2. **Defenders need to understand the threats.** Grid operators, infrastructure security teams, and turbine manufacturers benefit from understanding how number-theoretic tools could be used against their systems. Silence helps attackers more than defenders.

3. **Moral obligation.** If we propose technologies, we must also assess their misuse potential. Publishing only the positive applications would be intellectually dishonest and ethically incomplete.

2 Risk 1: Precision Harmonic Injection Attacks on Power Grids

2.1 The Positive Technology

The Möbius Harmonic Analyzer (MHA) decomposes grid harmonics into independent sources, enabling targeted active filtering.

2.2 The Dark Mirror

The same Möbius decomposition that identifies harmonic sources can be used to *synthesize* optimal harmonic injections. An attacker with a sufficiently powerful inverter connected to the grid could:

1. **Targeted transformer saturation**: Inject even-order harmonics (2nd, 4th) that cause DC offset in the transformer core flux, leading to half-cycle saturation. The Möbius decomposition identifies exactly which harmonic frequencies will bypass existing passive filters.

2. **Capacitor bank resonance amplification**: Identify the resonant frequency of grid capacitor banks ($f_r = 1/(2\pi\sqrt{LC})$) and inject power at that frequency. If f_r is near an integer harmonic, the arithmetic structure of the injection maximizes energy transfer.

3. **Protective relay confusion**: Craft a waveform whose zero-crossing pattern mimics a fault condition, causing protective relays to trip unnecessarily. The Möbius-optimal harmonic mixture minimizes the injected power needed to trigger a trip.

4. **Stealth**: By using Möbius inversion, the attacker can design an injection that appears as "natural" harmonic distortion (consistent with normal nonlinear load behavior) rather than an obvious attack signal.

2.3 Severity Assessment

Factor	Assessment
Novelty	Low — harmonic injection is a known attack vector
Precision gain	Medium — Möbius optimizes the injection
Stealth gain	**High** — attack looks like natural distortion
Physical access required	Grid connection (industrial inverter)
Detection difficulty	High if designed using Möbius structure
Potential impact	Localized transformer/capacitor damage

2.4 Defenses

1. **Deploy the MHA defensively**: The same tool that enables the attack also detects it. An MHA-equipped grid monitor can identify "unnatural" harmonic source patterns that don't match known load profiles.

2. **Harmonic injection limits**: IEEE 519 already sets limits on harmonic current injection. Enforce these at the point of common coupling with real-time monitoring.

3. **Anomaly detection**: Train ML models on the Möbius-decomposed harmonic spectrum to detect injections that deviate from historical patterns.

3 Risk 2: Resonance-Targeted Industrial Sabotage

3.1 The Positive Technology

The Arithmetic Vibration Monitor (AVM) uses compressed sensing with an arithmetic dictionary to detect incipient turbine faults.

3.2 The Dark Mirror

If arithmetic compressed sensing can detect subtle vibration signatures, it can also *design* them. An attacker with physical access to a turbine (or the ability to modulate its load) could:

1. **Resonance excitation**: Identify the natural frequencies of turbine blades via the Gram matrix eigenvalue structure, then apply periodic load variations at a blade natural frequency to induce resonant fatigue failure.

2. **Sub-harmonic injection**: Excite blade vibration at a frequency that is $1/k$ of a natural frequency (k prime), so the vibration builds gradually through nonlinear coupling rather than direct resonance. This is harder to detect because the excitation frequency is not near any monitored harmonic.

3. **Monitoring evasion**: Design the excitation signal to be orthogonal to the AVM's sensing dictionary, making it invisible to the monitoring system while still delivering destructive energy to the blade.

3.3 Severity Assessment

Factor	Assessment
Novelty	Medium — resonance attacks exist, arithmetic targeting is new
Precision gain	**High** — spectral analysis identifies exact modes
Stealth gain	**High** — sub-harmonic excitation evades monitoring
Physical access required	Load modulation or physical access to nacelle
Detection difficulty	Very high for sub-harmonic approach
Potential impact	Blade failure, turbine destruction, potential casualties

3.4 Defenses

1. **Broadband monitoring**: Monitor vibration across a wider frequency range than just the expected harmonics. Specifically monitor sub-harmonics at f_{nat}/p for prime p.

2. **Load pattern anomaly detection**: Flag unusual load variation patterns, especially periodic patterns at non-integer frequencies.

3. **Physical security**: Restrict physical access to turbine nacelles and control systems. This is the most effective defense against all vibration-based attacks.

4 Risk 3: Verified Instability Certificates

4.1 The Positive Technology

The Cathedral's framework produces machine-verified stability certificates for grid configurations: formal proofs that a Lyapunov function is positive definite.

4.2 The Dark Mirror

The same framework can produce machine-verified *instability* certificates: formal proofs that a specific perturbation sequence causes cascade failure.

1. **Optimal cascade sequences**: Given a grid model, find the minimum set of generator trips that causes voltage collapse. The Lean 4 verification ensures the attack plan is mathematically correct — no "wasted" disruptions.

2. **Stability boundary mapping**: Identify the exact operating conditions where the grid transitions from stable to unstable. An attacker who can push the grid to this boundary (e.g., by coordinating large loads) can trigger instability with minimal visible action.

3. **Targeted N-k analysis**: Current grid security standards require N-1 or N-2 contingency analysis (survive the loss of 1 or 2 components). A verified instability certificate could identify the specific N-3 or N-4 scenarios that current analysis misses.

4.3 Severity Assessment

Factor	Assessment
Novelty	**High** — verified attack plans are new
Precision gain	**Very High** — mathematically optimal attack
Stealth gain	Medium — the attack itself is visible
Access required	Grid topology and parameters (often semi-public)
Detection difficulty	Low (the cascade is observable)
Potential impact	**Regional blackout, economic damage**

This is the most concerning risk in this paper. The novelty is not in the attack (deliberate grid disruption is well-studied) but in the **verification**: a machine-checked proof that the attack *will* succeed, removing uncertainty from the attacker's planning.

4.4 Defenses

1. **Use the same tool defensively**: If the attacker can find verified instabilities, the grid operator should find them first. Deploy the same Lean 4 framework to systematically identify and remediate all N-3 and N-4 vulnerabilities.

2. **Protect grid models**: Limit public access to detailed grid topology and impedance data. Current CEII (Critical Energy Infrastructure Information) rules should be reviewed and strengthened.

3. **Dynamic reconfiguration**: Design grids that can rapidly change topology in response to detected instability, invalidating pre-computed attack plans.

4. **Adversarial stability margins**: Design with larger stability margins specifically in regions where N-3/N-4 analysis reveals verified instabilities.

5 Risk 4: Prime-Spaced Stealth and Jamming

5.1 The Positive Technology

Prime-spaced sensor arrays with Selberg beamforming weights offer broadband performance with predictable sidelobe control.

5.2 The Dark Mirror

The same array design, used in reverse, can produce electromagnetic emissions that are:

1. **Broadband noise with minimal power**: A prime-spaced jammer distributes its energy across frequencies without periodic structure, making it harder to filter. Conventional interference cancellation relies on identifying the jammer's spectral structure; a prime-spaced jammer has no periodic structure to identify.

2. **Difficult to geolocate**: The aperiodic radiation pattern of a prime-spaced array produces a complex spatial pattern that conventional direction-finding algorithms (designed for uniform or periodic arrays) may fail to resolve.

3. **Radar-evasive scattering**: A target coated in prime-spaced resonant elements would scatter radar returns across frequencies without the signature Bragg peaks of a periodic structure, reducing radar cross-section.

5.3 Severity Assessment

Factor	Assessment
Novelty	Low — aperiodic jamming/stealth are well-studied
Precision gain	Low — random spacing works comparably
Stealth gain	Low — prime spacing has marginal advantage
Access required	RF hardware (commercially available)
Detection difficulty	Medium
Potential impact	Communications disruption

Honest assessment: This is the weakest risk in the paper. Prime-spaced arrays are marginally better than random arrays for jamming/stealth purposes, and the advantage does not justify concern beyond existing aperiodic jamming threats.

5.4 Defenses

Existing anti-jamming techniques (spread spectrum, frequency hopping, adaptive beamforming) are effective against aperiodic jammers regardless of whether the aperiodicity comes from random or prime spacing.

6 Risk 5: Covert Broadband RF Surveillance

6.1 The Positive Technology

Prime-indexed rectennas harvest ambient RF energy across a wide bandwidth, enabling battery-free IoT sensors.

6.2 The Dark Mirror

A device that harvests broadband RF also *receives* broadband RF. A prime-indexed rectenna array disguised as an energy harvester could:

1. **Passive SIGINT**: Simultaneously monitor WiFi (2.4/5 GHz), cellular (700 MHz–3 GHz), Bluetooth (2.4 GHz), and IoT (900 MHz) without active emissions, making it undetectable by RF sweeps that look for transmitters.

2. **Self-powered**: The same harvested energy that powers the device also carries the intelligence. No battery means no maintenance visits, no supply chain to trace, and indefinite operational life.

3. **Plausible deniability**: If discovered, the device appears to be a legitimate energy harvester (IoT sensor, smart building component) rather than a surveillance device.

6.3 Severity Assessment

Factor	Assessment
Novelty	Medium — passive RF surveillance exists, self-powered is new
Precision gain	Low
Stealth gain	**High** — disguised as legitimate IoT hardware
Access required	Physical placement in target facility
Detection difficulty	**Very high** — no RF emissions to detect
Potential impact	Espionage, corporate/government intelligence

6.4 Defenses

1. **Physical inspection**: Regular RF environment audits that include physical inspection of all connected devices.
2. **Faraday shielding**: Sensitive facilities should control RF ingress/egress regardless of harvesting technology.
3. **Supply chain security**: Verify the provenance of all IoT/energy harvesting devices deployed in sensitive environments.

7 Risk Assessment Summary

Risk	Novelty	Severity	Feasibility	Primary Defense
Grid harmonic injection	Low	Medium	High	MHA defensive monitoring
Vibration sabotage	Medium	High	Medium	Broadband monitoring
Verified instability	**High**	**Very High**	Medium	Defensive verification
Stealth/jamming	Low	Low	High	Existing countermeasures
RF surveillance	Medium	Medium	Medium	Physical/supply chain

7.1 The Headline Risk

The single most concerning capability is **Risk 3: Verified Instability Certificates**. The ability to produce a *machine-checked proof* that a specific attack sequence will cause a grid cascade failure is qualitatively different from simulation-based attack planning. A simulation can be wrong; a verified certificate cannot (within its axioms).

The mitigation is symmetric: grid operators should use the same framework to find these vulnerabilities first. The race is between defensive and offensive verification, and the defender has the advantage of knowing their own system.

8 The Broader Dual-Use Question

None of the risks identified in this paper are created by the Cathedral. The mathematical tools (Fourier analysis, spectral theory, Möbius function, variational methods) have been publicly available for decades to centuries. What the Cathedral contributes is:

1. **Organization**: The Cathedral assembles these tools into a coherent framework. An attacker would otherwise need to independently recognize the connections between number theory and engineering.
2. **Verification**: The machine-verified proof methodology could, in principle, be applied to verify attack plans rather than proofs.
3. **Precision**: The number-theoretic structure (primes, Möbius function, Selberg weights) provides optimal or near-optimal parameters for certain engineering tasks, including adversarial ones.

The dual-use question is not whether to publish (the math is already public) but whether to *emphasize the connections* between number theory and infrastructure engineering. We believe the answer is yes, because:

> *The asymmetry in dual-use technology favors the defender.*

Defenders (grid operators, turbine manufacturers, infrastructure security teams) have domain knowledge, system access, and institutional resources. Attackers must operate covertly, with limited access and imperfect models. Providing defenders with a new mathematical framework strengthens them more than it strengthens attackers.

9 Recommendations

1. **Deploy the MHA defensively before it is used offensively.** The Möbius Harmonic Analyzer is the most immediately actionable tool in both the positive and negative applications. Grid operators should adopt it for monitoring before attackers adopt it for injection design.
2. **Extend N-k contingency analysis.** Use the Cathedral's verified stability framework to systematically identify N-3 and N-4 grid vulnerabilities. Remediate them before an adversary discovers them independently.
3. **Add sub-harmonic channels to vibration monitors.** Current turbine monitoring focuses on integer harmonics of the rotation frequency. Adding monitoring at f/p for small primes p would detect the sub-harmonic excitation attack described in Risk 2.
4. **Classify verified instability certificates.** Machine-checked proofs of grid instability should be treated as sensitive information (CEII) and protected accordingly.

5. **Publish this paper.** Suppressing dual-use analysis protects no one. The mathematical tools are public. The infrastructure is public. The only thing that can be private is the defender's understanding of their own vulnerabilities—and that understanding is strengthened, not weakened, by open analysis.

10 Conclusion

The Cathedral's mathematics is neutral. The Parseval Bridge connects time and frequency domains. The Möbius function decomposes multiplicative structure. The Rayleigh quotient bounds eigenvalues. These tools do not care whether they are used to protect a power grid or to attack one.

What matters is who learns to use them first.

This paper exists to ensure that the people who protect critical infrastructure understand the same mathematics that could threaten it. We believe that open, honest analysis of dual-use risks—even uncomfortable analysis—is a more effective defense than silence.

The Cathedral was built to illuminate. This paper points the light at the shadows.

A tool that can diagnose can also deceive.
A proof that certifies stability can also certify its absence.
The mathematics does not choose sides.
We must.

References

[1] IEEE, *IEEE Std 519-2022: Recommended Practice for Harmonic Control*, 2022.

[2] NERC, *Grid Security Guidelines: Critical Infrastructure Protection Standards CIP-002 through CIP-014*, 2023.

[3] R. Langner, *Stuxnet: Dissecting a Cyberwarfare Weapon*, IEEE Security & Privacy, 2011.

[4] FERC, *Critical Energy Infrastructure Information (CEII) Regulations*, 18 CFR §388.113.

[5] J.R. Gochanour, *What Could Be Built: Seven Engineering Frontiers*, 2026.

[6] J.R. Gochanour, *The Prime Number Grid: Energy Systems Applications*, 2026.